2023 Winter Qual

III.

$$\frac{\bar{y}_1 - \bar{y}_2 - (\mu_1 - \mu_2)}{\sigma\sqrt{n^{-1}}} \sim N(0,1)$$

$$\frac{\hat{n} S_p^2}{\sigma^2} = \frac{(n_1-1)S_1^2}{\sigma^2} + \frac{(n_2-1)S_2^2}{\sigma^2} = \chi^2(n_1-1) + \chi^2(n_2-1)$$

$$\sim \chi^2(n_1+n_2-2)$$

$$\Rightarrow \frac{\hat{n} S_p^2}{\sigma^2} / \hat{n} \sim \chi^2(n_1+n_2-2)/\hat{n}$$

$$\bar{y}_1, \bar{y}_2 \perp\!\!\!\perp S_1^2, S_2^2 \Longrightarrow \bar{y}_1 - \bar{y}_2 \perp\!\!\!\perp S_p^2$$

$$t = \frac{\dfrac{\bar{y}_1 - \bar{y}_2 - (\mu_1 - \mu_2)}{\sigma\sqrt{n^{-1}}}}{\sqrt{\dfrac{\hat{n} S_p^2}{\sigma^2} / \hat{n}}} = \frac{Z}{\sqrt{\chi^2(\hat{n})/\hat{n}}}$$

$$\sim t(n_1+n_2-2)$$

$$t^2 \sim F(1, \hat{n})$$

$\boxtimes$

$\boxed{3}$.

(a) $\hat{\beta} = (X^T X)^{-1} X^T Y$

$$\begin{pmatrix} 1^T 1 & \overset{=0}{1^T X_c} \\ \underset{=0}{X_c^T 1} & X_c^T X_c \end{pmatrix}^{-1} \begin{pmatrix} 1^T \\ X_c^T \end{pmatrix} Y = \begin{pmatrix} n 1^T \\ (X_c^T X_c)^{-1} X_c^T \end{pmatrix} Y$$

$$= \begin{pmatrix} n 1^T Y \\ \boxed{(X_c^T X_c)^{-1} X_c^T Y} = \hat{\beta}_1 \end{pmatrix}$$

$$\hat{\beta}_1 = (X_c^T X_c)^{-1} X_c^T Y = \hat{\beta}_c$$

$\blacksquare$

(b) $R^2 = \dfrac{SSR}{SST} = \dfrac{Y^T (H_X - H_1) Y}{Y^T (I - H_1) Y}$

$$\sum (y_k - \bar{y})^2 = ((I - H_1) Y)^T ((I - H_1) Y) = Y^T (I - H_1) Y$$

$$\hat{\beta}_c^T X_c^T X_c \hat{\beta}_c = Y^T H_{X_c} Y = Y^T (H_X - H_1) Y$$

$$H_X = X (X^T X)^{-1} X^T = (1 \quad X_c) \begin{pmatrix} 1^T 1 & 0 \\ 0 & X_c^T X_c \end{pmatrix}^{-1} \begin{pmatrix} 1^T \\ X_c^T \end{pmatrix}$$

$$= H_1 + H_{X_c} \rightsquigarrow H_{X_c} = H_X - H_1$$

So, $R^2 = \dfrac{\hat{\beta}_c^T X_c^T X_c \hat{\beta}_c}{\sum_{k=1}^{n} (y_k - \bar{y})^2}$

$$n\bar{y} = 1^T Y \Rightarrow n\bar{y}^2 = (1^T Y)^T (1^T 1)^{-1} (1^T Y) = Y^T H_1 Y$$

$$\frac{\hat{\beta}^T X^T Y - n\bar{y}^2}{Y^T Y - n\bar{y}^2} = \frac{Y^T(H_X - H_1)Y}{Y^T(I - H_1)Y} = \frac{SSR}{SST} = R^2$$

$\blacksquare$

(c) $$\cos\theta = \frac{Y^T \hat{Y}}{\|Y^T Y\| \, \|\hat{Y}^T \hat{Y}\|} = \frac{Y^T H_X Y}{\sqrt{Y^T Y}\sqrt{Y^T H_X Y}}$$

$$= \sqrt{\frac{Y^T H_X Y}{Y^T Y}} = R$$

$\blacksquare$